Ahmadyar Qureshi

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Education

PhD in Materials Science and Engineering (MSEN), Texas A&M University, College Station, Texas **May 2026 - December 2027**

- GPA: 3.60/4.0 | Expected graduation December 2027

MSc in Materials Science and Engineering (MSEN), Texas A&M University, College Station, Texas **January 2024 - May 2026**

- GPA: 3.60/4.00 | **Best MS Thesis Award**

BSc in Chemistry, Qatar University, Doha, Qatar **January 2020 – December 2023**

- GPA: 3.40/4.00 | Dean's List (1 year) | Vice-President's List (2 years)

Work Experience

National Corrosion and Materials Reliability Lab (NCMRL), ***Bryan, Texas*** **January 2024 – PRESENT**

- *Graduate Research Assistant - Corrosion & Materials Integrity*, **Texas A&M University**, College Station, Texas
- Evaluated corrosion and materials performance of sucker-rod steels under HPHT sour-service environments containing **100% CO_2, 100% H_2S, and 50/50 H_2S/CO_2**, using an integrated high-pressure testing and analysis setup
- Investigated hydrogen embrittlement susceptibility and residual-stress effects in additively manufactured 316LN stainless steel following electrochemical hydrogen charging (**50 mA/cm², 3 days**), correlating electrochemical measurements with material degradation
- Evaluated **additively manufactured Cu-Ni alloys** for corrosion and **stress-corrosion cracking (SCC)** susceptibility, examining the influence of **2 build orientations and temperatures up to 80 °C** in aggressive chloride environments on material integrity
- Tested carbon- and glass-fiber-reinforced polymer composites under pressurized hydrogen at **35 and 70 bar total pressure for 3–5 days** to determine degradation and suitability for hydrogen storage and transportation infrastructure
- Engineered Co-, Cu-, and CoCu-based electrocatalysts with telluride modifications for **hydrogen production via water electrolysis**, assessing **HER performance** through LSV, CV, Tafel analysis, EIS, and durability testing in **3 alkaline electrolyte environments**
- Conducted **mechanical and fracture testing** of metallic materials, including tensile, SSRT, hardness, CTOD/CMOD, and hydrogen embrittlement testing, to characterize mechanical integrity, cracking susceptibility, and failure behavior under service-relevant conditions
- **Executed proprietary industry-sponsored corrosion testing of α-HESA acid systems**, including electrochemical/materials performance assessment under application-relevant conditions
- Coordinated specimen fabrication and preparation via **EDM, waterjet cutting, machining, and controlled thermomechanical processing** for mechanical, corrosion, and materials qualification testing

SET Laboratories Inc., ***Intern Corrosion Scientist, Houston, Texas*** **May 2026 – August 2026**

- *Corrosion Scientist Intern – CCUS & Pipeline Integrity*
- Assessed **API 5L X52/X65 pipeline steels** under dense and supercritical CO_2 environments to quantify corrosion risks associated with CCUS transportation
- Designed and executed **50 °C/80 bar** autoclave testing under relevant H_2O and impurity conditions representative of CO_2 transportation streams
- Examined effects of **H_2O, O_2, SO_2, NO_2, and H_2S** impurities on uniform and localized corrosion behavior
- Developed experimental protocols for impurity-controlled dense/supercritical CO_2 testing, including fluid preparation, transfer, exposure, post-test characterization, and corrosion-rate assessment
- Generated materials-performance data supporting an **industry consortium** focused on pipeline integrity and CO_2 transportation

Center for Advanced Materials, ***Doha, Qatar*** **September 2021 – October 2023**

- *Undergraduate Research Assistant*, **Qatar University**
- Produced and validated **smart corrosion-protective coating systems** incorporating inhibitor-loaded mesoporous ceria and talc carriers into polymeric matrices for metallic substrate protection
- Formulated **pH-responsive inhibitor systems** using sodium benzoate and 8-hydroxyquinoline to provide active corrosion protection and structured inhibitor release
- Performed coating/material characterization using **EIS, salt-spray exposure, SEM/TEM, XRD, FTIR, and UV–Vis** to correlate coating chemistry, morphology, and barrier performance
- Optimized particle synthesis, inhibitor loading, and coating formulation, achieving approximately **10^9 Ω·cm² impedance** and substantially improved long-term corrosion resistance

Chiyoda Al Mana engineering LLC., ***Intern, Doha, Qatar*** **June 2023 – August 2023**

- Supported **brownfield LNG and wastewater-reuse engineering projects**, reviewing process modifications and existing facility constraints
- Interpreted and reviewed **PFDs and P&IDs** to understand equipment, piping, process streams, and proposed modifications
- Completed **hydraulic and depressurization calculations** supporting process engineering evaluations
- Prepared engineering documentation and process workflows supporting brownfield modifications and project deliverables

Qatar Environment and Energy Research Institute (QEERI) Education City, ***Intern, Doha, Qatar*** **June 2022 – August 2022**

- Engineered catalysts for **dry methane reforming** and received technical training in microstructural characterization techniques including SEM, TEM, and mass spectrometry as an Intern

Qatar Petrochemical Company (QAPCO), ***Intern, Doha, Qatar*** **June 2021 – August 2021**

- Acquired comprehensive knowledge of polyethylene manufacturing processes, plant operations, process equipment, utility systems, and operational safety through structured training across multiple production units

Undergraduate Research Experience Program, ***Qatar University, Doha, Qatar*** **January 2022 – December 2024**

- Fabricated **MXene**-based molecularly imprinted electrochemical sensors for trace chemical marker detection (UREP-28)

- Evaluated mineral content in desalinated water, crops, and dairy products through spectroscopic analysis (UREP-29)

Publications

1. Strategies to enhance the corrosion resistance of electrode materials in direct seawater electrolysis, **2026**, **A. Qureshi et al**., 10.1016/j.ijhydene.2026.157473
2. Additive Manufacturing of Glass Fiber-Reinforced Polypropylene and Its Behavior Under Controlled Hydrogen Exposure, **2026.** **A. Qureshi et al.**, https://doi.org/10.1002/pc.71564
3. Advances in photothermal catalysts for solar-driven hydrogen production, **2024**. ***A. Qureshi***, M. A. Wahab, A. Abdel-Wahab**,** H. Castaneda, and A. Abdala, https://doi.org/10.1016/j.ijhydene.2024.11.1242.
4. Advancing steel protection with Ceria@Talc-8-hydroxyquinoline modified Polyurethane coatings, **2024**. ***A. Qureshi***, S. Habib, Muhammad Faisal, Ramazan Kahraman, Elsadig Mahdi Ahmed, and R. A. Shakoor**,** https://doi.org/10.1016/j.porgcoat.2024.108446
5. Emerging trends in niobium, vanadium, and molybdenum based MXenes applications, **2024**, ***A. Qureshi*** et al., https://doi.org/10.1080/10408436.2022.2150124
6. TiO2-Mesoporous Ceria Carrier Modified with Sodium Benzoate: An Innovative Polyurethane Matrix for Enhanced Corrosion Protection of Steel, 2024. S. Habib, ***A. Qureshi,*** Samra Sajjad, Elsadig Mahdi Ahmed, and R. A. Shakoor, doi.org/10.1016/j.colsurfa.2024.13447
7. Corrosion inhibition performance of polyolefin smart self-healing composite coatings modified with ZnO@b-Cyclodextrin hybrid particles, **2022**, . S. Habib, ***A. Qureshi,*** R. A. Shakoor, Ramazan Kahraman, Noora Hamad Al-Qahtani, and Elsadig Mahdi Ahmed, doi.org/10.1016/j.jmrt.2022.10.148.
8. Modified halloysite nanotubes decorated with Ceria for synergistic corrosion inhibition of Polyolefin based smart composite coatings, **2023**, ***A. Qureshi,*** S. Habib, *Mudassir Nawaz*, , R. A. Shakoor, Ramazan Kahraman, and Elsadig Mahdi Ahmed, doi.org/10.1016/j.clay.2023.106827.
9. Functionalized nanoclay-based polymeric nanocomposites, **2024**, Sehrish Habib, Muddasir Nawaz*,* ***A.Qureshi***, RA Shakoor, Ramazan Kahraman, Elsadig Mahdi Ahmed, https://doi.org/10.1016/B978-0-443-18860-2.00009-8.
10. Picomolar or beyond limit of detection using molecularly imprinted polymer-based electrochemical sensors: A review, **2022**, Naheed Sidiq Shah, Vandana Thotathil, Shabi Abbas Zaidi, ***A.Qureshi***, https://doi.org/10.3390/bios12121107.

Conferences and Presentations

1. AMPP Annual Conference 2026 **March 2026**
A. Qureshi, Tanveer Ul Haq, Homero Castaneda "Corrosion-Resistant Electrocatalysts for Efficient Hydrogen Production via Direct Seawater Electrolysis," **Houston, Texas.**

2. American Chemical Society Spring 2023 National Conference **March 2023**
Ahmadyar Qureshi, Sehrish Habib, Muddasir Nawaz, R. A. Shakoor, Ramazan Kahraman, Elsadig Mahdi Ahmed, "Modified halloysite nanotubes decorated with Ceria for synergistic corrosion inhibition of Polyolefin based smart composite coatings", Poster presentation in ACS 2023 Crossroads of Chemistry, March 26-30, 2023, **Indianapolis, USA.**

3. American Chemical Society Fall 2023 National Conference **August 2023**
A. Qureshi, S. Habib, M. Nawaz, R. A. Shakoor, R. Kahraman, and E. M. Ahmed, "Corrosion inhibition performance of polyolefin smart self-healing composite coatings modified with ZnO@β-Cyclodextrin hybrid particles", Poster Presentation in ACS 2023 Harnessing the Power of Data, **San Francisco, California USA.**

4. Gulf Petrochemical and Chemical Association Competition **September 2022**
Sehrish Habib, Ahmadyar Qureshi, R. A. Shakoor, Ramazan Kahraman, Noora Hamad Al-Qahtani, Elsadig Mahdi Ahmed, "Smart Single Layer Polyolefin Coatings for Corrosion Protection of steel parts (S2Coat)", Oral presentation in Gulf Petrochemical and Chemical Association (GPCA) Research Innovation Competition held in **UAE.**

5. QU Creativity & Innovation for Engineering Solution Development Conference **April 2024**
A. Qureshi, S. Habib, M. Nawaz, R. A. Shakoor, R. Kahraman, and E. M. Ahmed, "Modified halloysite nanotubes decorated with Ceria for synergistic corrosion inhibition of Polyolefin based smart composite coatings,". 1st position in poster presentation in the QU Creativity & Innovation for Engineering Solution Development Conference, **Qatar University, Doha.**

6. Qatar University Annual Research Forum **October 2022**
Sehrish Habib, Ahmadyar Qureshi, R. A. Shakoor, Ramazan Kahraman, Noora Hamad Al-Qahtani, Elsadig Mahdi Ahmed, "Corrosion inhibition performance of polyolefin smart self-healing composite coatings modified with ZnO@β-Cyclodextrin hybrid particles", Qatar University Annual Research Forum (QUARFE), **Doha, Qatar.**

7. Qatar University Annual Research Forum **September 2022**
TiO2@GO nanocomposite: Preparation, characterization, and dye adsorption properties. Hayarunnisa Anwar, Bagmita Bhattacharyya, Ahmadyar Qureshi, Maimoona Mohamed, Hanan Sheikh, Sarya Muhannad Alfarwati, Rouzait Anwar Jarrar, Shabi Abbas Zaidi, Kishor Kumar Sadasivuni, Annual Research Forum & Exhibition 2022" (QUARFE'22) organized by **Qatar University**.

8. Qatar University Annual Research Forum **October 2022**
Ahmadyar Qureshi, 'Is FIFA World Cup 2022 Soccer Ball Sustainable? Chemistry point of view", Qatar University Research Forum.

Extracurricular Activities

1. Vice- President, American Chemical Society (ACS), **Qatar University Student Chapter** **September 2022 – February 2023**

- Coordinated with ACS national team to organize the 2022 MENA Conference and led outreach programs such as the 3rd and 4th edition of the chemistry festival to promote STEM engagement through interactive experiments and encourage youth to pursue careers in STEM.

2. Vice-President, **Qatar University e-Sports Club** **January 2021 – December 2024**

- Administered all managerial activities and developed meaningful collaborations with national and international partners.

3. Vice-Captain, **Qatar University Handball and Cricket Team.** **January 2021 – December 2024**

4. Co-chair, **MAROON Model United Nations (MUN) 2023 Conference.** **March 2023**

Technical Expertise

Corrosion & Integrity: HPHT Corrosion | Sour Service | CO_2/H_2S Corrosion | CCUS Pipeline Integrity | Hydrogen Embrittlement | SCC | Localized Corrosion | Protective Coatings
Electrochemistry: EIS | LPR | CPP | OCP/Ecorr | Potentiodynamic Polarization | Hydrogen Charging
HPHT Testing: Autoclave Systems | Dense/Supercritical CO_2 | Impurity-Controlled Testing | CO_2/H_2S Environments
Hydrogen & Energy: Hydrogen Materials | Water Electrolysis/HER | Electrocatalysis | Hydrogen Storage & Transportation
Materials & Characterization: Additive Manufacturing | SEM/EDS | EBSD | XRD | TDS | TEM | FTIR | Profilometry
Engineering: PFD/P&ID Interpretation | Hydraulic Calculations | Depressurization | Experimental/Test Protocol Development

References

1. Dr. Homero Castaneda
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